

Problem 1: Back Savers Backpack Production

- a. Decision Variables
 - a. $X_1 = \text{number of Collegiate backpacks produced per week}$
 - b. $X_2 = \text{number of Mini backpacks produced per week}$
- b. Objective Function
 - a. Each Collegiate backpack generates a profit of \$32 and each Mini backpack generates a profit of \$24.
 - b. Maximize $Z = 32X_1 + 24X_2$
 - c. Where: $Z = \text{total weekly profit}$
- c. Constraints
 - a. Nylon Fabric Constraint
 - i. Each Collegiate requires 3 square feet of nylon and each Mini requires 2 square feet.
 - ii. Only 5,000 square feet are available per week.
 - iii. Equation: $3X_1 + 2X_2 \leq 5000$
 - b. Labor Constraint
 - i. Each Collegiate requires 45 minutes of labor.
 - ii. Each Mini requires 40 minutes of labor.
 - iii. The company has:
 - 1. 35 workers x 40 hours = 1400 labor hours
 - iv. Convert to minutes:
 - 1. $1400 \times 60 = 84,000$ minutes
 - v. Therefore:
 - 1. $45X_1 + 40X_2 \leq 84000$
 - c. Sales Constraint
 - i. At most 1,000 Collegiates can be sold weekly:
 - 1. $X_1 \leq 1000$
 - ii. At most 1,200 Minis can be sold weekly:
 - 1. $X_2 \leq 1200$
 - d. Non-Negativity Constraint
 - i. $X_1 \geq 0$
 - ii. $X_2 \geq 0$
- d. Full Mathematical Formulation
 - a. Maximize $Z = 32X_1 + 24X_2$
 - b. Subject to:

- i. $3X_1 + 2X_2 \leq 5000$
- ii. $45X_1 + 40X_2 \leq 84000$
- iii. $X_1 \leq 1000$
- iv. $X_2 \leq 1200$
- v. $X_1, X_2 \geq 0$

Problem 2: Weigelt Corporation

a. Decision Variables

a. Plant 1

- i. X_{1L} = Large units produced at Plant 1
- ii. X_{1M} = Medium units produced at Plant 1
- iii. X_{1S} = Small units produced at Plant 1

b. Plant 2

- i. X_{2L} = Large units produced at Plant 2
- ii. X_{2M} = Medium units produced at Plant 2
- iii. X_{2S} = Small units produced at Plant 2

c. Plant 3

- i. X_{3L} = Large units produced at Plant 3
- ii. X_{3M} = Medium units produced at Plant 3
- iii. X_{3S} = Small units produced at Plant 3

d. P = common proportion of excess capacity utilized by each plant

b. Objective Function

a. Profit per unit:

- i. Large = \$420
- ii. Medium = \$360
- iii. Small = \$300

b. Maximize Formula

$$Z = 420(X_{1L} + X_{2L} + X_{3L}) + 360(X_{1M} + X_{2M} + X_{3M}) + 300(X_{1S} + X_{2S} + X_{3S})$$

c. Constraints

a. Capacity Constraints

- i. Plant 1: $750P = (X_{1L} + X_{1M} + X_{1S})$
- ii. Plant 2: $900P = (X_{2L} + X_{2M} + X_{2S})$
- iii. Plant 3: $450P = (X_{3L} + X_{3M} + X_{3S})$
- iv. These constraints ensure each plant uses the same percentage of its excess capacity.

b. Storage Constraints

- i. Plant 1: $20X_{1L} + 15X_{1M} + 12X_{1S} \leq 13000$
- ii. Plant 2: $20X_{2L} + 15X_{2M} + 12X_{2S} \leq 12000$
- iii. Plant 3: $20X_{3L} + 15X_{3M} + 12X_{3S} \leq 5000$

c. Sales Constraints

- i. Large: $X_{1L} + X_{2L} + X_{3L} \leq 900$
- ii. Medium: $X_{1M} + X_{2M} + X_{3M} \leq 1200$
- iii. Small: $X_{1S} + X_{2S} + X_{3S} \leq 750$

d. Proportional Constraint

- i. $0 \leq P \leq 1$

e. Non-negativity Constraint

- i. $X_{ij} \geq 0$

1. Note: for all plants i and sizes j

d. Full Mathematical Formulation

- a. Maximize: $Z = 420(X_{1L} + X_{2L} + X_{3L}) + 360(X_{1M} + X_{2M} + X_{3M}) + 300(X_{1S} + X_{2S} + X_{3S})$

b. Subject To:

- i. $750P = (X_{1L} + X_{1M} + X_{1S})$
- ii. $900P = (X_{2L} + X_{2M} + X_{2S})$
- iii. $450P = (X_{3L} + X_{3M} + X_{3S})$
- iv. $20X_{1L} + 15X_{1M} + 12X_{1S} \leq 13000$
- v. $20X_{2L} + 15X_{2M} + 12X_{2S} \leq 12000$
- vi. $20X_{3L} + 15X_{3M} + 12X_{3S} \leq 5000$
- vii. $X_{1L} + X_{2L} + X_{3L} \leq 900$
- viii. $X_{1M} + X_{2M} + X_{3M} \leq 1200$
- ix. $X_{1S} + X_{2S} + X_{3S} \leq 750$
- x. $0 \leq P \leq 1$
- xi. $X_{ij} \geq 0$